

Dual Drive VoltSense ESC (Electronic Speed Controller)

Technical Specifications

Parameter	Specification
Product Name	Dual Drive VoltSense ESC
Board Dimensions	60 mm (W) × 40 mm (H)
PCB Thickness	1.6 mm FR-4 (Typical)
Board Layers	2 Layer
Input Voltage	Up to 27 V DC
Motor Channels	2 (Dual Drive)
Maximum Output Current	43 A per channel
Control Interface	RC PWM / ESP32
Wireless Communication	Bluetooth (ESP32)
Battery Monitoring	Real-time Voltage Sensing
Motor Feedback	Motor Voltage & Current Monitoring
Protection Features	Under-voltage, Over-current, Reverse Polarity, Fault Indication
Indicators	Power LED, Status LEDs, Buzzer
Applications	Combat Robots, AGVs, RC Vehicles, Robotics, Industrial Automation

Notes

This document summarizes the public technical specifications of the Dual Drive VoltSense ESC. Detailed firmware, PCB design files, schematics, and manufacturing data are intentionally omitted to protect the associated intellectual property and published patent.